

ABHISHEK LOHAR

SOFTWARE DEVELOPMENT • AI/ML SYSTEMS • FULL-STACK • BACKEND ARCHITECTURE

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PROFESSIONAL SUMMARY

Highly motivated Computer Science and Engineering student at VIT-AP with a strong focus on Artificial Intelligence, Deep Learning, and backend development. Experienced in building AI-powered applications, RAG pipelines, full-stack systems, and scalable software solutions. Passionate about open-source development and solving real-world problems through technology.

CORE TECHNICAL COMPETENCIES

Programming Languages:	Java, Python, SQL, C++, JavaScript, HTML/CSS
AI, ML & Vector Search:	LLMs, RAG, Embeddings, LangChain, ChromaDB, Ollama, OpenAI, OpenCV, Data Science
Backend & Web Systems:	FastAPI, React, PostgreSQL, Redis, Apache Kafka, RESTful APIs, Microservices
DevOps, Testing & Tools:	Git, GitHub Actions CI/CD, PyPI Package Deployment, pytest (160+ tests), Raspberry Pi
Core CS Fundamentals:	Data Structures & Algorithms, Object-Oriented Design (OOD), DBMS, Operating Systems, Computer Networks

EDUCATION

Vellore Institute of Technology-AP (VIT-AP) — Amaravati, India	2023 – 2027
<i>B.Tech in Computer Science and Engineering — CGPA: 8.33 / 10.0</i>	
Vidya Bharati Chinmaya Vidyalaya (CBSE) — Jamshedpur, India	2022
<i>Higher Secondary Education (Class XII) — Aggregate: 79.8%</i>	
Vidya Bharati Chinmaya Vidyalaya (CBSE) — Jamshedpur, India	2020
<i>Secondary Education (Class X) — Aggregate: 85.8%</i>	

FEATURED ENGINEERING PROJECTS

local-persona-memory — Production Open-Source Python Package (PyPI) | Python, LangChain, ChromaDB, Ollama, RAG, pytest, GitHub Actions

- Architected and published open-source Python package on **PyPI** (pip install local-persona-memory), giving local LLMs autonomous long-term persistent memory with 100% data privacy and zero cloud dependence.
- Designed resilient RAG pipeline incorporating ChromaDB vector embeddings, chunked PDF document ingestion, sub-second semantic search, and dynamic conversation recall hooks. Shipped with **160 passing tests** and GitHub Actions CI/CD.

Fact Knowledge Layer — Document Intelligence System | Python, FastAPI, OpenAI, ChromaDB, SQLite, React

- Built enterprise document intelligence pipeline extracting atomic, auditable facts from financial PDFs with value, unit, time, scope, source quote, and page-level provenance using GPT-4o mini and Pydantic.
- Engineered two-stage reconciliation engine using ChromaDB semantic retrieval to prune irrelevant fact pairs before GPT-4o reasoning, reducing potential $O(N^2)$ comparisons by ~98%.

Distributed URL Shortener & Analytics Engine | FastAPI, PostgreSQL, Redis, Apache Kafka, Microservices Architecture

- Developed decoupled microservices backend featuring isolated Shorten, Redirect, and Analytics services coordinated behind API Gateway.
- Configured **Redis in-memory caching** for sub-5ms redirects with **PostgreSQL** persistence, integrated **Apache Kafka** for asynchronous click metrics streaming.

CERTIFICATIONS, HONORS & AWARDS

- **Oracle Cloud Infrastructure (OCI) Certified AI Foundations Associate:** Enterprise AI & cloud paradigms.
- **Finalist, Engineering Clinics 2025:** Recognized among top engineering innovators for practical hardware-software solutions.
- **High-Performance Coding (HPC) Certification (IAMNEO):** Algorithmic problem solving and runtime execution optimization.
- **Infosys Springboard — Introduction to Data Science (May 2026):** Data wrangling, analytics, and statistical modeling.
- **Hashgraph Developer Course & Hack2Skill GDG Hackathons:** Distributed ledger technology & collaborative hackathon prototyping.

LANGUAGES & INTERESTS

Languages: English (Professional Working Proficiency), Hindi (Native) | **Interests:** Open Source Development, LLM Autonomous Agents, Edge AI, Distributed Systems